

Roll No.

TBT-601

B. TECH. (BIOTECHNOLOGY) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

ANIMAL BIOTECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is animal cell culture ? Explain in detail. (CO1)

OR

(b) What is the difference between transport and growth medium ? Give some examples. (CO1)

2. (a) Elaborate the method of trypsinization. (CO1)

OR

(b) Explain Serum media and its importance. (CO1)

3. (a) What is cryopreservation ? Give the method for preserve germplasm. (CO1/CO2)

OR

(b) Write an essay on transgenic animals and its importance. (CO1/CO2)

P. T. O.

(2)

4. (a) Give the examples of transgenic fishes. (CO2)

OR

- (b) Give a short note on Enviro pig. (CO2)

5. (a) What is the role of biotechnology in livestock production ? (CO2)

OR

- (b) Elaborate the method of superovulation and embryo transfer. (CO2)

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TBT-602

B. TECH. (BIOTECHNOLOGY) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

PLANT BIOTECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is culture media ? Explain the components of MS media. (CO1)

OR

(b) Explain the principle of Plant tissue culture. Write advantages and disadvantages of PTC. (CO1)

2. (a) What is somatic embryo genesis ? Explain with suitable diagram.

(CO1/CO2)

OR

(b) What is somaclonal variation ? Explain the molecular basis of somaclonal variation. (CO1/CO2)

3. (a) Explain the method of synthetic seed production. How is it different from natural seed ? (CO2)

P. T. O.

(2)

OR

- (b) What is synthetic seed ? How is it different from natural seed ? (CO2)
4. (a) What do you understand by protoplast fusion ? Explain its application in Plant tissue culture. (CO2)

OR

- (b) Discuss the role of PTC in conservation of threatened plant species. (CO2)
5. (a) What is cryopreservation ? Explain the method of cryopreservation. (CO3)

OR

- (b) What is nitrogen fixation ? Explain the molecular basis of nitrogen fixation. (CO3)

Roll No.

TBT-603

B. TECH. (BIOTECHNOLOGY) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

BIOSEPARATION TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Describe *five* types of Bioproducts obtained in the Fermentation Industry. Give example of each type of product. (CO1)

OR

- (b) Discuss various Unit operations applied in Downstream processing and their basis. (CO1)

2. (a) What are various problems faced in downstream processing of Bioprocess products ? (CO1)

OR

- (b) Describe the historical development of Bioprocess industry since pre 1900 era. (CO1)

3. (a) Explain Freundlich Isotherm and Langmuir Isotherm in detail. What are the disadvantages of Freundlich Isotherm ? (CO1)

P. T. O.

(2)

OR

(b) Describe principle of Centrifugation. Discuss the equipment, working and applications of Disc Bowl Centrifuge. (CO2)

4. (a) What is Solvent extraction ? Describe advantages and problems associated with solvent extraction. (CO2)

OR

(b) Write short notes on the following : (CO2)

(i) Salting out using Ammonium Sulphate

(ii) Isoelectric precipitation and Cohn equation

5. (a) Explain principle of Chromatography. How can we increase resolution of chromatographic column ? (CO3)

OR

(b) Describe AGAROSE GEL electrophoresis in detail. How is it different from Pulsed field gel electrophoresis. (CO3)

Roll No.

TBT-604(a)

B. TECH. (BIOTECHNOLOGY) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

BIOSIMILAR TECHNOLOGIES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are the differences between a generic and a biosimilar ? (CO1)

OR

- (b) What are the differences between a biologic and a biosimilar ? (CO1)

2. (a) Describe goal, scope, structure and components of biosimilar technology regulatory guideline of India. (CO1)

OR

- (b) "Biosimilar market growth driven by increasing development of biologics." Comment. (CO1)

3. (a) Describe the prerequisites for biosimilar development. (CO1/CO4)

OR

- (b) Describe the factors to be considered for manipulating immunogenicity of biosimilar. (CO1/CO4)

P. T. O.

(2)

4. (a) Define types of bioequivalence : average, population, individual.

(CO4)

OR

- (b) Which applications are found missing from the Orange Book ? (CO4)

5. (a) What role does immunogenicity play in biosimilars ? (CO4)

OR

- (b) What are the difference between average and population bioequivalence ? (CO4)

Roll No.

TBT-604(b)

B. TECH. (BIOTECHNOLOGY) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

BIOSENSOR AND BIOELECTRONICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss the different types of electrochemical biosensors and their applications. (CO1/CO2)

OR

- (b) Describe the applications of biosensors in biomedical sciences.

(CO1/CO2)

2. (a) What are the optical biosensors ? Discuss the principle and applications of one optical biosensor. (CO4/CO2)

OR

- (b) What are the whole cells based biosensors ? Discuss the working mechanism of one whole cell based biosensor. (CO4/CO2)

3. (a) What do you understand by a transducer ? Explain the role of a transducer in biosensor. (CO3)

P. T. O.

TCS-471

V1th Sem

MID SEMESTER EXAMINATION, April, 2023

Time : 1½ Hours

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

1. (a) Why is Statistical Data Analysis using R important in Digital era ?

OR

(CO1)

(i) 3, 5, 2, 6, 5, 9, 5, 2, 8, 6

(ii) 51.6, 48.7, 50.3, 49.5, 48.9

OR

(C03)

P. T. O.

(2)

3. (a) A fair die is tossed twice. Find the probability of getting a 4, 5, or 6 on the first toss and a 1, 2, 3, or 4 on the second toss. (CO3)

OR

- (b) Ram and Shyam play 12 games of chess, of which 6 are won by A, 4 are won by B, and 2 end in a draw. They agree to play a match consisting of 3 games. Find the probability that (i) A wins all 3 games, (ii) 2 games end in a draw, (iii) A and B win alternately and (iv) B wins at least 1 game. (CO3)

4. (a) How is standard packages different from command packages in R ? Illustrate the steps for Installing Extra Packages for Windows Users.

(CO2)

OR

- (b) Explain the motivations behind using R as compared to Python. (CO2)

5. (a) Create two objects data1 and data2 of weekdays and months in a year respectively. Write a function in R to save these objects with proper extension, remove these objects, check these objects and recovery of these objects in workspace. (CO3)

OR

- (b) You have a dataset named test.csv having multiple entries with two columns the data is to be read along with the header, in the second scenario there is delimiter space in same file. How to read these files and handle missing values in R ? (CO3)